

ISLE OF PORTLAND, United Kingdom

WMO: 038570

Lat: 50.5218N

Lon: 2.4573W

Elev: 171

StdP: 14.61

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.2	33.5	23.7	18.0	34.1	26.3	20.4	35.8	45.7	47.1	41.6	46.9	17.4	50	0.782

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	70.4	63.2	68.1	62.3	66.4	61.4	64.6	68.1	63.6	66.5	62.7	65.2	11.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.3	87.5	65.8	62.4	84.7	64.7	61.6	82.3	64.0	29.7	68.4	29.0	66.6	28.3	65.4	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
39.8	34.6	31.5	DB	29.5	75.3	3.1	3.5	27.3	77.9	25.5	79.9	23.7	81.9	21.5	84.4
			WB	27.6	66.8	3.0	1.7	25.4	68.0	23.7	69.0	22.0	70.0	19.8	71.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	44.6	44.0	45.5	48.7	53.2	57.9	61.7	62.6	60.2	56.0	50.3	46.2	(6)
	DBStd	7.69	5.00	4.43	3.77	3.72	3.53	2.91	2.82	2.37	2.66	3.82	4.57	5.39	(7)
	HDD50	749	172	170	141	63	10	0	0	0	0	6	52	135	(8)
	HDD65	4538	634	589	604	489	366	214	110	83	144	279	442	584	(9)
	CDD50	1703	3	1	2	24	109	238	362	390	307	192	59	16	(10)
	CDD65	16	0	0	0	0	0	1	7	7	1	0	0	0	(11)
	CDH74	15	0	0	0	0	0	2	7	6	0	0	0	0	(12)
	CDH80	1	0	0	0	0	0	0	0	1	0	0	0	0	(13)
(14) Wind	WSAvg	16.5	19.9	18.2	16.5	14.4	14.2	13.5	13.5	14.2	15.9	18.5	19.0	20.0	(14)
(15) Precipitation	PrecAvg	32.9	3.8	2.8	2.4	2.1	2.0	2.0	1.8	2.3	2.5	3.7	3.9	3.7	(15)
	PrecMax	44.4	7.6	6.5	5.1	5.4	4.4	4.8	4.8	5.1	5.1	5.6	8.1	6.8	(16)
	PrecMin	26.3	0.7	0.4	0.1	0.1	0.3	0.3	0.2	0.4	0.5	1.3	0.9	1.1	(17)
	PrecStd	4.7	1.7	1.5	1.0	1.3	1.1	1.1	1.1	1.3	1.3	1.3	1.7	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	52.9	55.3	62.9	67.4	71.1	74.6	73.7	69.7	64.5	59.5	55.8	(19)
		MCWB	51.8	50.6	49.1	55.0	59.6	62.9	63.7	66.1	63.0	60.7	57.8	54.1	(20)
	2%	DB	52.5	51.2	52.4	58.4	63.1	66.8	70.5	70.0	67.0	62.7	58.0	54.9	(21)
		MCWB	51.0	49.6	49.2	52.0	57.2	60.3	62.9	63.8	61.8	59.7	56.4	53.4	(22)
	5%	DB	51.5	50.2	51.0	55.4	60.3	64.3	68.1	67.7	65.4	61.6	57.2	53.9	(23)
		MCWB	50.1	48.6	48.6	50.8	55.9	59.2	62.1	62.6	60.7	58.9	55.5	52.4	(24)
	10%	DB	50.6	49.3	50.0	53.3	58.2	62.3	66.2	66.2	64.1	60.7	56.2	52.9	(25)
		MCWB	49.0	47.7	47.9	49.7	54.7	58.2	61.3	61.6	60.0	58.2	54.2	51.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.8	51.7	51.3	56.1	61.7	63.7	65.5	67.3	64.7	62.1	58.6	55.2	(27)
		MCDB	53.0	52.0	52.8	61.6	65.4	69.0	71.0	72.1	67.7	63.3	58.9	55.3	(28)
	2%	WB	51.6	50.3	50.3	53.1	58.3	61.5	63.9	65.1	63.4	61.0	57.2	54.0	(29)
		MCDB	52.1	50.8	51.4	56.5	61.6	65.4	68.3	68.3	65.7	62.0	57.6	54.5	(30)
	5%	WB	50.5	49.3	49.4	51.7	56.7	60.1	62.9	63.9	62.4	60.0	56.0	53.0	(31)
		MCDB	51.2	49.9	50.5	54.1	59.4	63.2	66.6	66.4	64.2	61.1	56.8	53.7	(32)
	10%	WB	49.6	48.3	48.6	50.6	55.4	59.0	62.0	62.9	61.5	59.0	54.7	51.7	(33)
		MCDB	50.4	49.0	49.7	52.6	57.7	61.5	65.2	65.0	63.3	60.3	55.9	52.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	5.7	6.4	7.4	7.5	7.4	7.3	7.0	7.3	6.0	5.7	5.5	(35)
		MCDBR	4.8	5.3	6.8	9.9	10.1	10.1	10.1	8.8	8.3	5.8	5.1	4.9	(36)
	5% WB	MCWBR	6.1	5.5	5.6	6.8	7.0	6.8	6.5	6.0	5.8	5.3	6.1	6.4	(37)
		MCWBR	4.9	5.0	5.8	7.6	8.9	8.6	8.5	7.4	6.8	5.1	4.9	4.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.333	0.348	0.382	0.396	0.399	0.392	0.387	0.380	0.371	0.360	0.333	0.323	(40)
		taud	2.385	2.397	2.306	2.295	2.333	2.383	2.402	2.442	2.452	2.449	2.434	2.361	(41)
	Ebn at Noon		219	248	258	268	272	275	274	271	260	240	218	202	(42)
		Edh at Noon	21	26	34	38	38	36	35	32	29	24	19	18	(43)
(44) All-Sky Solar Radiation	RadAvg		260	497	888	1399	1689	1832	1741	1458	1100	614	329	207	(44)
	RadStd		23	54	112	123	89	180	125	114	69	44	32	25	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (63 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page